

SMART INDIA HACKATHON 2026



RASTA AI

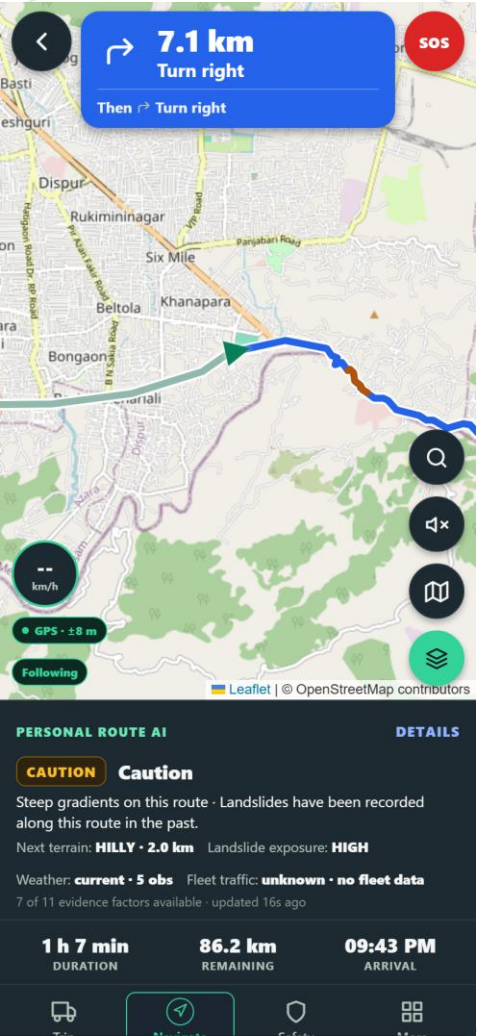
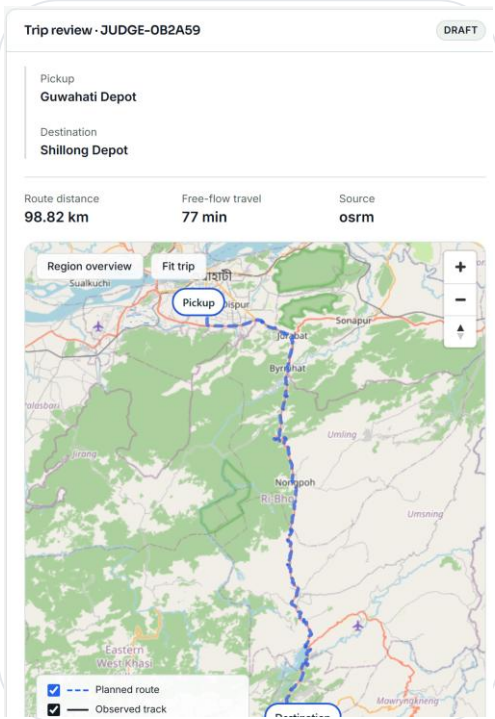
Route Intelligence for Essential-Supply Logistics in North Eastern India

SIH26002

Problem Statement ID
MDoNER · Smart Automation · Software

PROBLEM STATEMENT ID	SIH26002
PROBLEM STATEMENT TITLE	AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)
THEME	Smart Automation
PS CATEGORY	Software
TEAM ID	17 (internal group no.)
TEAM NAME	NER-AI LOGISTICS

End-to-end working prototype · physical-phone validated · live demo corridor Guwahati → Shillong



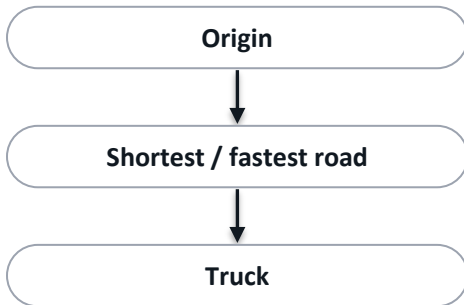
Manager route review · Driver navigation — real screens

PROPOSED SOLUTION

NOT THE SHORTEST ROAD. THE ROAD THAT IS OPERATIONALLY USABLE NOW.

NER corridors: mountain terrain · monsoon · landslides · floods · road disruption · weak connectivity · stale evidence. **A shortest-path router still sends the truck.**

A NORMAL ROUTER

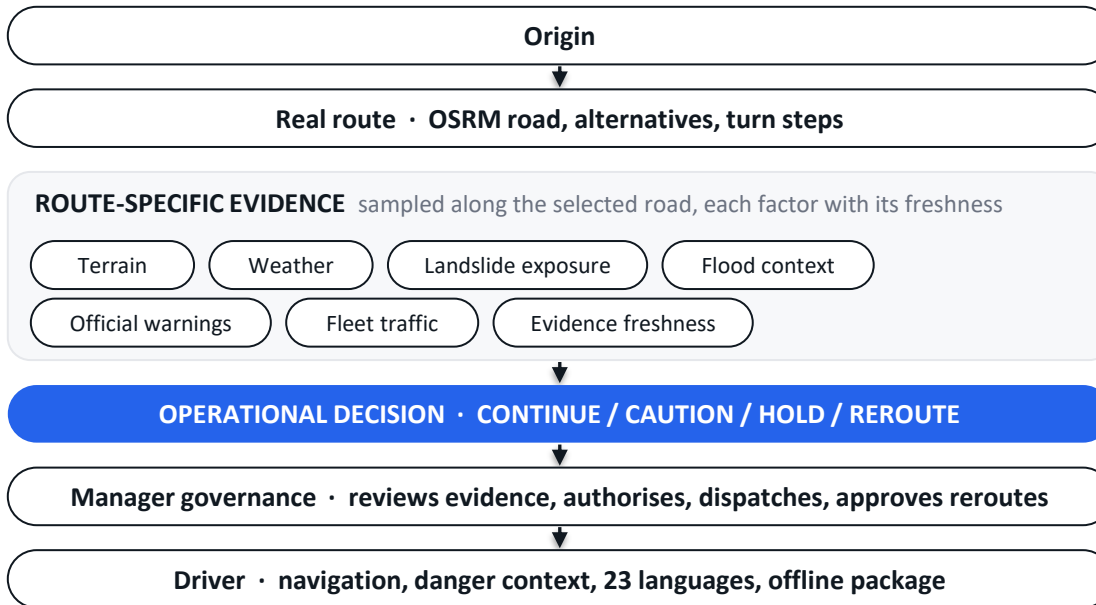


No terrain, weather, hazard or freshness check.

No governance. No answer when data is missing.

UNKNOWN ≠ SAFE

RASTA AI



The system refuses to fabricate certainty: missing or stale evidence is marked UNKNOWN, never counted as safe, and the route goes to an authorised reviewer.

REQUIRES REVIEW

Hazard evidence is incomplete or elevated. An authorised reviewer must review this route before selection. Refresh conditions after review.

Landslide data is not available - no source is connected. · Steep gradients on this route · Landslides have been recorded along this route in the past. · The landslide inventory ends in 2017 - recent years are not covered. · River levels near this route are close to their recent average. · No official disaster alert names a district on this route right now. · No fleet truck has driven this road in the last fifteen minutes, so traffic is unknown.

Unavailable: Fuel range, Landslide, Road quality, Truck restrictions

Evidence coverage · Weather AVAILABLE · Terrain AVAILABLE · Warnings AVAILABLE · Traffic UNKNOWN

TERRAIN

0.5 KM STEEP

49-1436 m · 1975 m climb · steepest 10.3%
flat 61.8 km · rolling 29.6 km · hilly 7.0 km · steep 0.5 km
Copernicus DEM GLO-90 via Open-Meteo + SRTM 30 m via OpenTopoData · 100% of 198 samples answered

LANDSLIDE HISTORY

HIGH EXPOSURE

22 recorded landslides within 5 km of the road, nearest 0.4 km
6 more nearby placed too imprecisely to count · HIGH is three or more; MODERATE is one or two
NASA Global Landslide Catalog · inventory 2007-2017 — aged, recent years not covered

RIVER LEVELS

NORMAL

Discharge at 5 river cells along the corridor, highest 1.0× its own 30-day mean · GloFAS via Open-Meteo, 2026-09-14. A level, not a flood forecast and not a road-closure claim.

OFFICIAL ALERTS

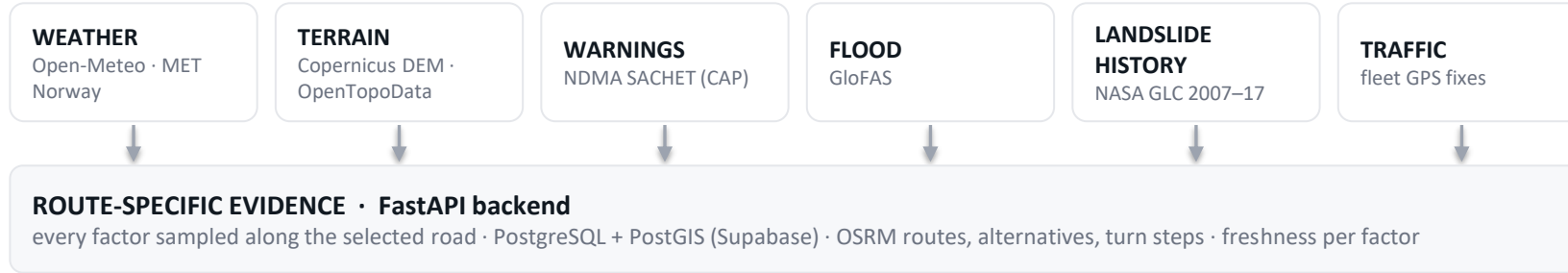
NONE ON CORRIDOR

NDMA SACHET CAP feed, 99 alerts nationwide at 8:32:18 PM · placed by district name (Dispur, East Khasi Hills, Kamrup Metropolitan, Myllem, Ri-Bhoi, Umling, Umsning) · 2 more active elsewhere in the corridor states, not placeable on this road.

Manager · Check conditions (real screen)

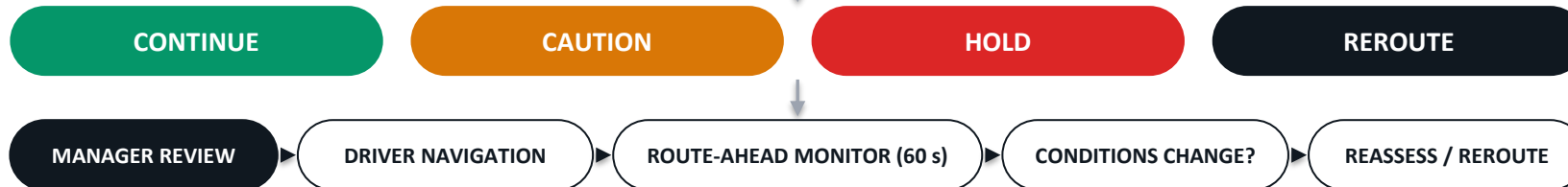
TECHNICAL APPROACH

VERIFIED DATA · provider adapters with health + freshness



DETERMINISTIC SAFETY POLICY

11 factors · reason codes · UNKNOWN ≠ SAFE · reviewer authorisation when hazard data is missing



Decision loop: conditions are re-scored every 60 s along the road ahead; a material change goes back through the policy, and a reroute is proposed to the manager — never applied silently.

CLIENTS Manager console — React + TypeScript, manager accounts only · Driver app — React Native / Expo, Android; one login, server-decided role, offline trip package

STACK FastAPI · PostgreSQL / PostGIS (Supabase) · React · React Native / Expo · OSRM · OpenStreetMap · Gemini

EVIDENCE Open-Meteo / MET Norway · NDMA SACHET · GloFAS · NASA historical landslides · OpenTopoData / Copernicus DEM

WHO DECIDES — AI IS SUPPORTING

DETERMINISTIC SAFETY ENGINE

makes every operational decision

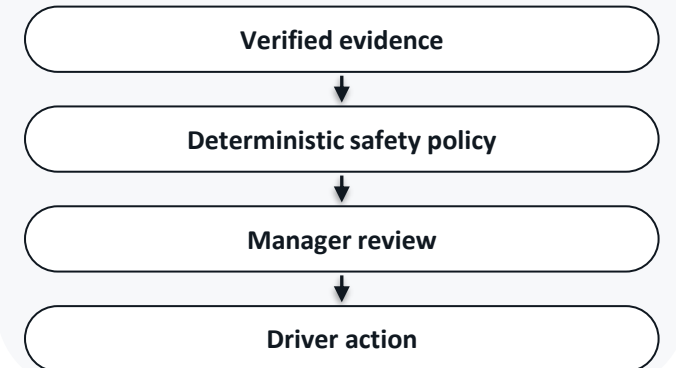
GEMINI / OPENROUTER (LLM)

explains the decision and assists in the driver's language — never decides

EXPERIMENTAL ML

research channel only; kept outside routing by the safety-validation gate

GOVERNANCE CHAIN



NOT A CONCEPT — A PHYSICAL, END-TO-END WORKING PROTOTYPE

MANAGER ROUTE REVIEW

DISPATCH

DRIVER ACCEPT + TRUCK CHECK

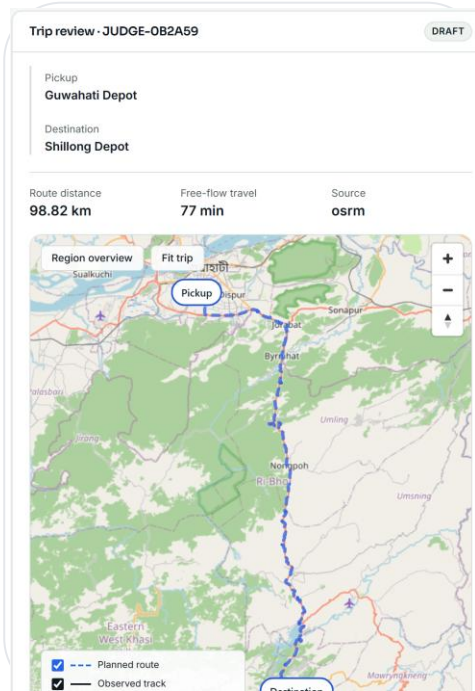
NAVIGATION

OFF-ROUTE

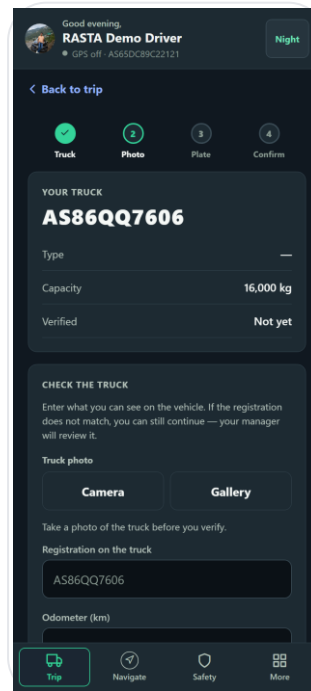
REROUTE

MANAGER APPROVAL

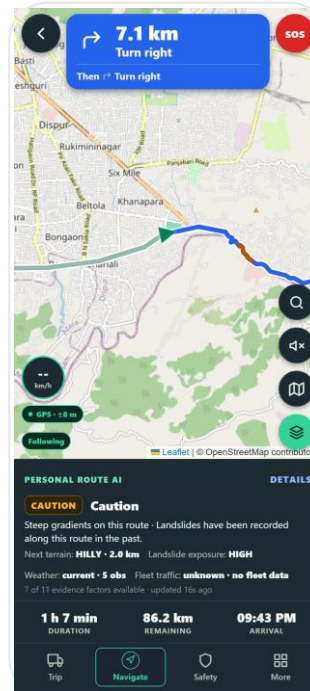
DELIVERY



Manager · route review



Driver · truck check



Driver · navigation

PROOF

Physical Android phone · hosted backend · real roads

12/12

judge-flow steps on the phone: dispatch → accept → truck check → GPS navigation → real off-route → reroute → manager approval → stops → delivery

9/9

role steps: driver ↔ manager sign-in on one APK, no data leak

1138+

backend tests · 621+ driver tests · 170+ manager tests

PHYSICAL ANDROID · CERTIFIED

Canonical demo Guwahati → Shillong · ≈ 98.8 km real route

KNOWN LIMITS → MITIGATION Render cold start → warm /health before the demo · provider rate limits → cached evidence + explicit UNKNOWN · background push needs Firebase config → in-app alerts already work · experimental ML → outside routing until the gate passes

IMPACT AND BENEFITS

RASTA AI does not ask only “Which road is shortest?” It asks “**Can this truck reliably use this corridor now?**”

FOOD • MEDICINE • FUEL • RELIEF SUPPLIES — essential logistics for hill communities, decided on evidence

WHO BENEFITS

FLEET MANAGERS

Route usability and uncertainty visible before dispatch; auditable decisions.

DRIVERS

Safer navigation, danger context, emergency support, 23 languages.

REMOTE COMMUNITIES

More resilient access to food, medicine, fuel and relief supplies.

GOVERNMENT / OPERATIONS

Auditable route evidence and disruption visibility per corridor.

DEMONSTRATED TODAY

23 selectable languages, explicit fallback status

11 route factors in every decision

60 s route-ahead monitor while driving

12/12 judge-flow steps certified on a physical phone

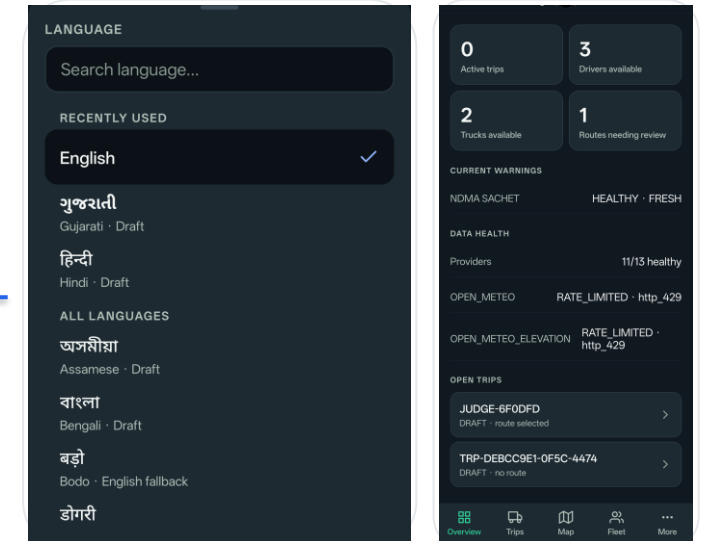
100 % of reroutes reviewed by a human

0 unknown factors counted as safe

HUMAN GOVERNANCE — AI HAS NO UNCONTROLLED AUTHORITY



No invented percentages: impact is stated as who gains what, backed by the working system.



Driver • 23 languages

Manager • mobile

Real screens, APK 1.0.18. One login, server-decided role: a manager gets a mobile fleet view, a driver gets navigation. Each language shows its status — Verified, Draft or English fallback — so nobody is misled.

RESEARCH / EVIDENCE SOURCES BEHIND THE ROUTE DECISION

NASA Global Landslide Catalog

historical landslide exposure; inventory 2007–2017, ≤ 5 km accuracy filter

NDMA SACHET (CAP feed)

official warnings, polled with freshness

GloFAS — Copernicus Emergency Management

flood context: river discharge vs 30-day mean

Open-Meteo · MET Norway

weather forecast along the corridor; ERA5-Land archive for research

Copernicus DEM GLO-90 · OpenTopoData

terrain profile, climb, steep segments

OpenStreetMap · OSRM

roads, route alternatives, turn steps

MDoNER problem statement SIH26002

AI-based smart logistics and accessibility intelligence for the NER

REFERENCES

gpm.nasa.gov/landslides (NASA GLC) · sachet.ndma.gov.in · global-flood.emergency.copernicus.eu (GloFAS) · open-meteo.com · api.met.no · opentopodata.org · Copernicus DEM · project-osrm.org · openstreetmap.org · github.com/nxtlucifer/ner-ai-logistics (repository, tests, model registry, error analysis)

EXPERIMENTAL LANDSLIDE RESEARCH · ML-ready architecture, model not in production

95 %

RECALL

NER geographic holdout (154 events)

50 %

FALSE-POSITIVE RATE

too high for production

SAFETY GATE · NOT DEPLOYED

trained India-wide, validated on the untouched NER holdout

TRAIN

HELD-OUT TEST

FALSE POSITIVES TOO HIGH

SAFETY GATE

NOT DEPLOYED

VALIDATION BEFORE AUTOMATION. The model is trained and geographically validated, but the safety-validation gate keeps it outside production until it meets the required geographic reliability and false-positive thresholds. Route decisions today use verified evidence and deterministic policy only.

Technical: precision 0.32 · PR-AUC 0.55 · ROC-AUC 0.83 · Brier 0.12 · rule baseline recall 0.25 @ FPR 0.03 · logistic regression on NASA GLC events + ERA5-Land rain + DEM slope · full error analysis in [docs/HAZARD_ERROR_ANALYSIS.md](#)